CARLO CAVICCHIA

PERSONAL DATA

✉ cavicchia@ese.eur.nl

Personal Page ◇ Institutional Page ◇ Google Scholar ◇ ORCiD ◇ Research Gate

CURRENT POSITION

Assistant Professor (Universitair Docent 1, UD1) of Statistics *December 2024 - Present*
Econometric Institute, Erasmus School of Economics, Erasmus Rotterdam University, Rotterdam

EDUCATION

Masterclass Supervising PhD Research *December 2024*
Erasmus University Rotterdam, Rotterdam

Dutch University Teaching Qualification UTQ-BKO *January 2022 - June 2022*
Risbo, Erasmus University Rotterdam, Rotterdam

Ph.D. in Statistics *November 2016 - February 2020*
Department of Statistical Sciences, University of Rome La Sapienza, Rome
Thesis title: *Hierarchical latent variable models for dimensionality reduction: an application on composite indicators*; Supervisor: Prof. M. Vichi

MSc Degree in Statistics and Decision Science *November 2013 - January 2016*
University of Rome La Sapienza, Rome 110/110 with honors

BSc Degree in Statistics *November 2009 - July 2013*
University of Rome La Sapienza, Rome 105/110

QUALIFICATIONS

National Qualification for Associate Professor (Italy, ASN II fascia), Statistics (SC 13/D1)

RESEARCH AND TEACHING INTERESTS

Hierarchical Models	Non-parametric Statistics	Data Science
Latent Variable Models	Composite Indicators	Unsupervised Classification

RESEARCH EXPERIENCES

Research visit *May 2026*
Department of Economics, Management and Institutions, Federico II University of Naples, Naples
Guest of Prof. P. Sarnacchiaro

Research visit *November 2024*
Department of Economics, Management and Institutions, Federico II University of Naples, Naples
Guest of Prof. P. Sarnacchiaro

Research visit *November 2019 – December 2019*
Econometric Institute, Erasmus School of Economics, Erasmus Rotterdam University, Rotterdam
Guest of Prof. P.J.F. Groenen

Research fellowship *May 2019 – April 2020*
Department of Law and Economics, University of Rome UnitelmaSapienza, Rome
Supervisor: Prof. P. Sarnacchiaro

Researcher on a project *March 2016 – November 2018*
Department of Statistical Sciences, University of Rome La Sapienza, Rome
Title: *Development of a procedure to detect and manage outliers for big data*
Supervisor: Prof. M. Vichi and ENEA

Researcher on a project *March 2016 – January 2017*
Department of Statistical Sciences, University of Rome La Sapienza, Rome
Title: *Construction of a hierarchical non-negative and disjoint composite indicator for a set of quantitative variables*
Supervisor: Prof. M. Vichi

AWARDS AND GRANTS

SSH Open Competition XS (49838.- EUR) *May 2026*
Nederlandse Organisatie voor Wetenschappelijk Onderzoek, The Hague, Netherlands
IFCS Chikio Hayashi Award (1000.- USD) *July 2024*
International Federation of Classification Societies, San Jose, Costa Rica
ESE Starter grant (300000.- EUR) *September 2023*
Erasmus School of Economics, Erasmus Rotterdam University, Rotterdam, The Netherlands
Research Grant for Junior Researchers (1270.- EUR) *August 2017*
University of Rome La Sapienza, Rome, Italy
Best report of Stats under the stars[3] *June 2017*
Italian Statistical Society (SIS), Florence, Italy
Ph.D. scholarship (3 years) *August 2016*
University of Rome La Sapienza, Rome, Italy

FEATURED PUBLICATIONS

van de Velden, M., A. Iodice D'Enza, A. Markos, and C. Cavicchia: *Unbiased mixed-variable distance*. In: *Journal of Computational and Graphical Statistics* (2026). DOI: 10.1080/10618600.2026.2680181.

Willemsen, R., C. Cavicchia, W. van den Heuvel, and M. van de Velden: *An Exact Solution Approach for Hierarchical Clustering*. In: *INFORMS Journal on Computing* 38.2 (2026), pp. 447–462. DOI: 10.1287/ijoc.2024.0903.

Bottazzi Schenone, M., C. Cavicchia, M. Vichi, and G. Zaccaria: *Ultrametric Factor Analysis for building hierarchies of reliable and unidimensional latent concepts*. In: *Psychometrika* 90.2 (2025), pp. 779–798. DOI: 10.1017/psy.2025.6.

van de Velden, M., A. Iodice D'Enza, A. Markos, and C. Cavicchia: *A general framework for implementing distances for categorical variables*. In: *Pattern Recognition* 153 (2024), p. 110547. DOI: https://doi.org/10.1016/j.patcog.2024.110547.

Cavicchia, C., M. Vichi, and G. Zaccaria: *Parsimonious Ultrametric Gaussian Mixture Models*. In: *Statistics and Computing* 34 (2024), p. 108. DOI: 10.1007/s11222-024-10405-9.

Cavicchia, C. and P. Sarnacchiaro: *A new hierarchical composite indicator model for ranking the top 20 European football teams*. In: *Quality and Quantity* 58.5 (2024), pp. 4033–4051. DOI: 10.1007/s11135-024-01842-9.

Cavicchia, C., P. Sarnacchiaro, M. Vichi, and G. Zaccaria: *A model-based ultrametric composite indicator for studying waste management in Italian municipalities*. In: *Computational Statistics* 39.1 (2024), pp. 21–50. DOI: 10.1007/s00180-023-01333-9.

Cavicchia, C., M. Vichi, and G. Zaccaria: *Hierarchical Disjoint Principal Component Analysis*. In: *Advances in Statistical Analysis* 107.3 (2023), pp. 537–574. DOI: 10.1007/s10182-022-00458-4.

Vichi, M., C. Cavicchia, and P.J.F. Groenen: *Hierarchical Means Clustering*. In: *Journal of Classification* 39.3 (2022), pp. 553–577. DOI: 10.1007/s00357-022-09419-7.

Cavicchia, C., M. Vichi, and G. Zaccaria: *Gaussian Mixture Model with an extended ultrametric covariance structure*. In: *Advances in Data Analysis and Classification* 16.2 (2022), pp. 399–427. DOI: 10.1007/s11634-021-00488-x.

Cavicchia, C. and M. Vichi: *Second-order disjoint factor analysis*. In: *Psychometrika* 87.1 (2022), pp. 289–309. DOI: 10.1007/s11336-021-09799-6.

Cavicchia, C. and P. Sarnacchiaro: *A multi-group higher-order factor analysis for studying the gender-effect in teacher job satisfaction*. In: *METRON* 80.1 (2022), pp. 23–38. DOI: 10.1007/s40300-021-00204-x.

Cavicchia, C. and P. Sarnacchiaro: *The effect of a new e-tivity on students' performance and satisfaction in an online course*. In: *Statistica Applicata - Italian Journal of Applied Statistics* 33.2 (2021), pp. 163–175. DOI: 10.26398/IJAS.0033-009.

Cavicchia, C., P. Sarnacchiaro, and M. Vichi: *A composite indicator for the waste management in the EU via hierarchical disjoint non-negative factor analysis*. In: *Socio-Economic Planning Sciences* 73 (2021), p. 100832. DOI: 10.1016/j.seps.2020.100832.

Cavicchia, C. and M. Vichi: *Statistical model-based composite indicators for tracking coherent policy conclusions*. In: *Social Indicators Research* 156.2 (2021), pp. 449–479. DOI: 10.1007/s11205-020-02318-7.

Cavicchia, C., M. Vichi, and G. Zaccaria: *The ultrametric correlation matrix for modelling hierarchical latent concepts*. In: *Advances in Data Analysis and Classification* 14.4 (2020), pp. 837–853. DOI: 10.1007/s11634-020-00400-z.

INVITED TALKS AND SEMINARS

IFCS 2026 - 19th conference of the IFCS *July 2026*
Milan, Italy
Seminar "Clustering Beyond Algorithms: Structure, Models, and Optimization" *May 2026*
DEMI Department - Federico II University of Naples
ISI 2025 - 65th International Statistical Institute World Statistics Congress *October 2025*
The Hague, Netherlands
Seminar "Distances for mixed data: issues, solutions and a new KNN approach" *November 2024*
DISES Department - Federico II University of Naples
IFCS 2024 - 18th Conference of the IFCS *July 2024*
San José, Costa Rica
SIS 2024 - 52nd Scientific meeting of the Italian Statistical Society *June 2024*
Bari, Italy
Seminar "Convex clustering of mixed numerical and categorical data" *May 2024*
DIETI Department - Federico II University of Naples
Workshop on "Statistical Computing and Robust Inference for High Dimensional Data" *December 2023*
Taipei, Taiwan
ISI 2023 - 64th International Statistical Institute World Statistics Congress *July 2023*
Ottawa, Canada
DSSV-ECDA 2023 - Joint conference of Data Science, Statistics & Visualisation and the European Conference on Data Analysis *July 2023*
Antwerp, Belgium
SIS 2023 - Statistical Learning, Sustainability and Impact Evaluation *June 2023*
Ancona, Italy
Seminar "Convex clustering of mixed numerical and categorical data" *April 2023*
Statistical Sciences Department - Sapienza University (Virtual)

CMStatistics 2022 - 15th International Conference of the ERCIM WG on Computational and Methodological Statistics *December 2022*
London, UK (Virtual)
IFCS 2022 - 17th Conference of the IFCS *July 2022*
Porto, Portugal
EcoSta 2022 - 5th International Conference on Econometrics and Statistics *June 2022*
Kyoto, Japan (Virtual) **Seminar series "Introduction to Statistics"** *February 2022*
Hospitality Management - Federico II University of Naples (Virtual)
CLADAG 2021 - 13th Scientific Meeting of Classification and Data Analysis Group *September 2021*
Florence, Italy (Virtual)
ISI 2021 - 63rd International Statistical Institute World Statistics Congress "Statistics and Data Science for a Better World" *July 2021*
Virtual conference
EconJobMarket *December 2019*
Rotterdam, Netherlands
ASA 2019 Scientific Conference on Statistics for Healt and Well-Being *September 2019*
Brescia, Italy
CLADAG 2019 - 12th Scientific Meeting of Classification and Data Analysis Group *September 2019*
Cassino, Italy
SIS 2018 - 49th Scientific meeting of the Italian Statistical Society *June 2018*
Palermo, Italy
Workshop on "Selection and Synthesis of Indicators for Sustainable Development Goals: Methodologies and Applications" *October 2017*
Rome, Italy

SUPERVISION

PhD (Leon Czarnecki) *2026 - Present*
Erasmus School of Economics, Erasmus University Rotterdam, Rotterdam
Econometrics and Operation Research
Research Internship (Alexia Chirita) *2025 - Present*
Erasmus School of Economics, Erasmus University Rotterdam, Rotterdam
Econometrics and Operation Research
BSc Honours Class thesis (Arianna Giuffra and Beatriz Polak) *2025*
Erasmus School of Economics, Erasmus University Rotterdam, Rotterdam
International Bachelor Economics and Business Economics
Research Internship (Leon Czarnecki) *2025 - Present*
Erasmus School of Economics, Erasmus University Rotterdam, Rotterdam
Research MSc Business Data Science
PhD (Irene Ariante) *2025 - Present*
Department of Economics, Management and Institutions, University of Naples Federico II, Naples
Statistics and Management
PhD (Marten Soer) *2024 - Present*
Erasmus School of Economics, Erasmus University Rotterdam, Rotterdam
Econometrics and Operation Research
Research Internship (Aleksandr Zhuravlev) *2024 - 2025*
Erasmus School of Economics, Erasmus University Rotterdam, Rotterdam
Research MSc Business Data Science
MSc theses (3) *2021 - Present*
Erasmus School of Economics, Erasmus University Rotterdam, Rotterdam
MSc Econometrics and Management Science, Econometrics

MSc theses (17) *2021 - Present*
Erasmus School of Economics, Erasmus University Rotterdam, Rotterdam
MSc Econometrics and Management Science, Business Analytics and Quantitative Marketing

MSc theses (20) *2021 - Present*
Erasmus School of Economics, Erasmus University Rotterdam, Rotterdam
MSc Economics and Business, Data Science and Marketing Analytics

BSc theses (33) *2020 - Present*
Erasmus School of Economics, Erasmus University Rotterdam, Rotterdam
BSc Econometrics and Operational Research

TEACHING AS COORDINATOR OR LECTURER

Seminar Data Science for Marketing Analytics *2025 - Present*
Erasmus School of Economics, Erasmus University Rotterdam, Rotterdam
MSc Economics and Business, Data Science and Marketing Analytics

Applied Statistics 2 *2022 - Present*
Erasmus School of Economics, Erasmus University Rotterdam, Rotterdam
BSc Economics and Business Economics

Toegepaste Statistiek 2 *2022 - Present*
Erasmus School of Economics, Erasmus University Rotterdam, Rotterdam
BSc Economics and Business Economics

Seminar Case Studies in Business Analytics and Quantitative Marketing *2022 - 2024*
Erasmus School of Economics, Erasmus University Rotterdam, Rotterdam
MSc Econometrics and Management Science, Business Analytics and Quantitative Marketing

Seminar in Business Analytics and Quantitative Marketing *2021*
Erasmus School of Economics, Erasmus University Rotterdam, Rotterdam
BSc Econometrics and Operational Research

Seminar Data Science for Marketing Analytics *2020 - 2022*
Erasmus School of Economics, Erasmus University Rotterdam, Rotterdam
MSc Economics and Business, Data Science and Marketing Analytics

Statistics *2019*
School of Economics, Tor Vergata University of Rome, Rome
Master of Science in European Economy and Business Law

EDITORIAL SERVICES

Associate editor of "vISIon" *2025 - Present*

Co-guest-editor of the special issue "Theoretical and Computational Advances in Classification, Clustering, Pattern Recognition and Data Analysis " in METRON *2025*

Associate editor of "Statistica Applicata – The Italian Journal of Applied Statistics" *2025 - Present*

Article Editor for Sage Open *2024 - Present*

Co-guest-editor of the special issue "Advances in statistical modelling, methods and applications" in METRON *2022*

Co-editor of "Book of Short Papers of SIS 2022" *2022*

Reviewer for a number of scientific journals in Statistics *2018 - Present*
– in alphabetic order – Advances in Data Analysis and Classification; Annals of Operations Research; AStA Advances in Statistical Analysis; Behaviormetrika; BMC Psychology; Computational Optimization and Applications; Computational Statistics; Computational Statistics and Data Analysis; Educational and Psychological Measurement; International Journal of Data Science and Analytics; Journal of Classification; Journal of Computational Social Science; Journal of Data Science, Statistics and Visualisation; Journal of Infrastructure, Policy and Development; Journal of Statistical Computation

and Simulation; Journal of the Royal Statistical Society Series B; METRON; Operational Research; PLOS ONE; Quality and Quantity, SAGE Open; Scientific Reports; SN Business & Economics; Socio-Economic Planning Sciences; SoftwareX; Statistica; Statistical Methods and Applications; Statistics and Computing

ACADEMIC SERVICES

Member of the scientific programme committee of IES 2027 *2027*
Organizer of an invited session the 5th ISI Regional Statistical Conference *2026*
Member of the scientific programme committee of the 5th ISI Regional Statistical Conference *2026*
Member of the organizing committee of the 2nd Annual SSH-Breed Sectorplan Conference *2026*
Member of the scientific board of CLADAG *2025 - Present*
Member of the committee of IASC data analysis competition *2025*
Organizer of an invited session of CLADAG-VOC 2025 *2025*
Organizer of an invited session of IES 2025 *2025*
Member of the scientific programme committee of CLADAG-VOC 2025 *2025*
Member of the scientific programme committee of IES 2025 *2025*
IASC data analysis competition officer *2023 - Present*
Member of the executive board of IASC *2023 - Present*
Member of the committee of IASC data analysis competition *2023*
Organizer of an invited session of CLADAG 2023 *2023*
Member of the scientific programme committee of CLADAG 2023 *2023*
Member of the scientific programme committee of IES 2023 *2023*
Organizer of an invited session of SIS 2023 *2023*
Responsible for the "Sustainability" topic within SVQS *2022 - Present*
Organizer of an invited session of CMStatistics 2022 *2022*
Co-organizer for the Econometrics internal seminars at Econometric Institute, Erasmus University Rotterdam *2022 - Present*
Organizer of an invited session of ECDA 2022 *2022*
Member of the scientific programme committee of HackTheGene 2022 *2022*
Member of the committee of the prize "Premio TStat Training" *2022*
National coordinator of y-SIS *2022*
Member of the scientific programme committee of SIS 2022 *2022*
Member of the scientific programme committee of IES 2022 *2022*
Member of the scientific board of y-SIS *2021 - 2022*
Member of the organizing committee of DSSV-ECDA 2021 *2021*
Member of the organizing committee of DSSV 2020 *2020*
Member of the admission committee of II level master: "Quantitative methods and models for decision support", University of Rome UnitelmaSapienza *2020*
Organizer of an invited session of SIS 2020 *2020*

MEMBERSHIPS

International Statistical Institute (ISI - Elected member) *2025*
Multi-dimensional Data Visualisation (MuViSU) *2025*
Erasmus Research Institute of Management (ERIM - Associate member) *2023*
Vereniging voor Ordinatie en Classificatie (VOC) *2022*
International Association Statistical Computing (IASC) *2020*
Statistica per la Valutazione e la Qualità nei Servizi (SVQS) *2019*
Classification and Data Analysis Group (CLADAG) *2019*

young SIS (y-SIS) *2016*
Società Italiana di Statistica - Italian Statistical Society (SIS) *2016*

WORK EXPERIENCES

Consultant as Lead Statistician *April - December 2026*
Project: *Construction of a composite indicator for monitoring the implementation of laws and policies related to abortion*, World Health Organization, Geneva, Switzerland
Consultant as Lead Statistician *October - December 2019*
Project: *Survey in Zanzibar on community socio-economic conditions*, ARS Progetti S.P.A., Rome, Italy **Consultant as Supervisor** *July - October 2019*
Project: *Survey in Zanzibar on environmental cleanliness, public service perception and tourism competitiveness*, NGO Fondazione ACRA, Rome, Italy
Consultant as Lead Statistician *June - August 2018*
Project: *Survey within the framework of the "Consultancy Services to support community engagement and awareness within the Urban Municipal Council"*, NGO Fondazione ACRA, Zanzibar, Tanzania

LANGUAGES

Italian Native language
English Speaking: C1 Writing: C2

EXTRA-CURRICULAR

- President and Co-Founder "ASD Valcanneto Futsal" (Cerveteri, 2018 - 2021)
- Attended the 1st International Summer School on Classification and Data Analysis (Rimini, 2017)
- Attended a Erasmus semester at University College of Dublin (Dublin, 2012)